

printing, multi-busbar designs, and gradual adoption of alternative metallisation approaches, have demonstrated the potential to reduce silver usage by 20–40% over time as these technologies are validated and scaled.

The world is moving towards cleaner energy sources and is hence scaling up solar power generation. Even as technology advancement is reducing silver usage, it is outweighed by the scale at which solar manufacturing is rising globally.

“Sustained elevation in metal prices does put pressure on the pace and economics of scaling up domestic manufacturing, particularly for new capacity additions. While cost-escalation clauses provide partial protection, pass-through is neither immediate nor complete, resulting in short-term margin compression,” says Mathur.

According to GREW Solar, its integrated manufacturing approach and scale-driven efficiencies allow them to absorb short-term fluctuations while main-

recycling initiatives. At the same time, strong domestic demand, supportive policy frameworks, and long-term competitiveness of solar energy continue to underpin capacity expansion plans.

Some industry players with integrated operations say these cycles act as catalysts for building resilient, future-ready manufacturing ecosystems. “We remain confident that as supply chains stabilise and efficiencies deepen, the industry will continue to scale up sustainably while maintaining stable pricing and strong growth fundamentals,” says Thadani.

Analysts, however, believe renewable players should be able to offset the rise in metal prices owing to the cost escalation clause. There is some headroom as far as margins are concerned, they say.

Copper prices have also shown variability, reinforcing the importance of diversified sourcing and prudent procurement strategies. In response, manufacturers are increasingly focusing on supplier diver-

SOLAR INDUSTRY PLAYERS REMAIN HOPEFUL AND SAY THE PRICES ARE A SHORT-TERM CHALLENGE... THEY ARE ACCELERATING INNOVATION, MATERIAL EFFICIENCY

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taining high quality, performance reliability, and timely deliveries.

“These measures help us navigate commodity cycles responsibly while continuing to support customers and project developers with dependable solutions,” says Thadani.

WAY FORWARD

The next chapter in India’s clean energy narrative begs another question: How will companies pivot to sustainable production methods to mitigate the risks of fluctuating commodity cycles?

Built-in cost escalation clauses are expected to shield them. Solar industry players remain hopeful and say the high prices are only a short-term challenge. Instead, they are accelerating innovation, material efficiency, and process optimisation.

Manufacturers are increasingly focused on reducing dependency on expensive inputs through technology improvements, alternative designs, and enhanced

sification, longer-term procurement arrangements, and greater process flexibility to manage material cost exposure.

“For India, as solar manufacturing capacity continues to expand, the emphasis must remain on long-term resilience rather than short-term cost absorption. Continued investment in technology, supply-chain partnerships, and selective upstream integration will be important to manage commodity volatility and strengthening competitiveness in the global solar manufacturing ecosystem,” says Jackson’s Chanana.

Clean energy led by solar remains firmly at the centre of India’s growth and energy transition strategy, which means high demand for key metals despite technology advancements. Companies have already factored in these requirements and are on track to absorb the metal rally risk. **BT**

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